

Geometric Series

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1 Geometric Series

Prove that:

$$\sum_{i=1}^n a \cdot r^{i-1} = \frac{a \cdot (1 - r^{n+1})}{1 - r}$$

Proof:

Define S_n such that:

$$S_n = a + ar + ar^2 + \dots + ar^{n-1} + ar^n$$

Note that

$$r \cdot S_n = ar + ar^2 + \dots + ar^{n-1} + ar^n + ar^{n+1}$$

Such that

$$\begin{aligned} S_n - r \cdot S_n &= a - ar^{n+1} \\ \Rightarrow S_n &= \frac{a \cdot (1 - r^{n+1})}{1 - r} \end{aligned}$$

If $|r| < 1$:

Then

$$\sum_{i=1}^{\infty} a \cdot r^{i-1} = \lim_{n \rightarrow +\infty} \frac{a \cdot (1 - r^{n+1})}{1 - r} = \frac{a}{1 - r}$$

Note that by taking derivatives of both sides and a bit of reindexing:

$$\begin{aligned} \sum_{i=0}^{\infty} \frac{d}{dr} a \cdot r^i &= \sum_{i=0}^{\infty} i \cdot a \cdot r^{i-1} = 0 + \sum_{i=1}^{\infty} i \cdot a \cdot r^{i-1} = \frac{a}{(1 - r)^2} \\ \Rightarrow \sum_{i=1}^{\infty} i \cdot r^i &= \frac{r}{(1 - r)^2} \text{ when } |r| < 1 \end{aligned}$$